

MECH1634 -Gripper Design and Palletizing

Emmanuel Muhiga & Ibrahim Adem

MECH1634

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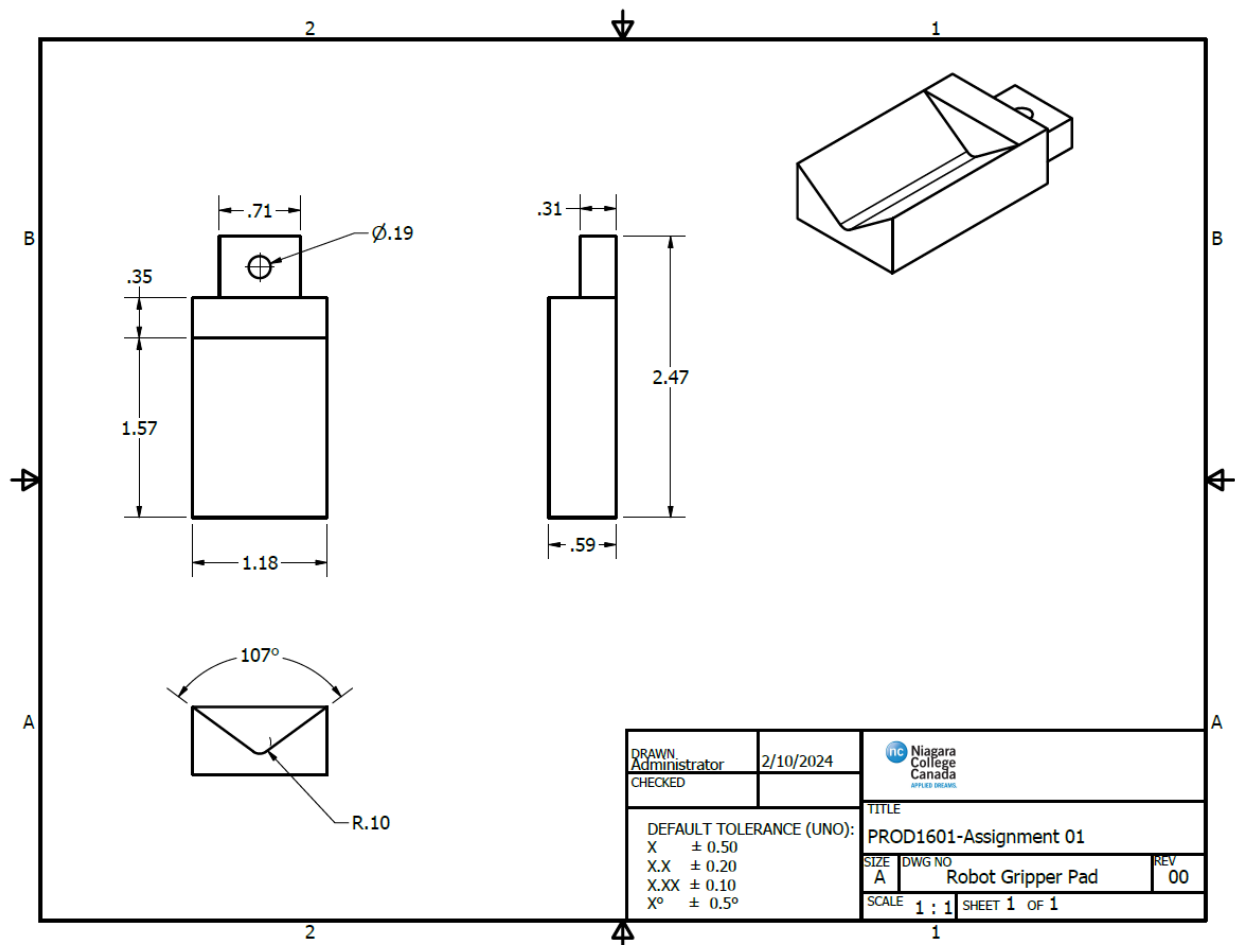
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1. Description of Operations

The program does load and unload operations. Unload pallet: The robot picks a cylinder from pallet and place on conveyor; Load Pallet: The robot picks cylinder from conveyor and place on pallet. The operator can choose which cylinder the robot picks up from the pallet by specifying its position on the array. The pallet is structured in an array, with 5 positions in the X direction with corresponding 4 positions in the Y direction.

2. Gripper finger fabrication



3. Point Location Table

Name	Description
pCylinderBase	Target of first cylinder in array {1,1}. All other targets associated with the pallet are offsets for this target.
pConveySim	Target of cylinder on the conveyor. All other targets associated with the conveyor are offsets for this target.

4. I/O Listing

Input		Output	
Name	Description	Name	Description
diGripClosed	Verify Gripper Closed	doGripActuate	Toggles between gripper close and open
diGripOpen	Verify Gripper Closed		

5. Observations and Errata

Overall, the program was successful. The robot can pick parts from the pallets and places the part on conveyor and vice versa. The robot gripper pads designed could not grip on cylinder when grippers are in a closed state. Another gripper finger was used as shown in the video.

6. Demonstration Video

<https://youtu.be/MBQvpd7QQ7k>

7. ABB RAPID Program

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MODULE Module1
! Lab Assignment 1 - Gripper Design and palletizing
! Emmanuel Muhiga & Ibrahim Adem
! Jan 27, 2024
! Description: Program to load or unload the pallet and conveyor

CONST robtargt pCylinderBase_10:=[12.7,25.4,-4.47],[0,-0.707106781,0.707106781,0],[0,0,0,0],[9E+09,9E+09,9E+09,9E+09,9E+09,9E+09];
CONST robtargt pConveySim_10:=[19.05,36.665,7.688],[0,1,0,0],[0,0,0,0],[9E+09,9E+09,9E+09,9E+09,9E+09,9E+09];

VAR robtargt pRBPickArray{5,4};
VAR num x:=0;
VAR num y:=0;
VAR bool bFirstPass:=TRUE;
VAR num nOperation:=0;

PROC main()
! Entry point
IF bFirstPass THEN
! Initialization code for first pass of program
TPERase;
rRelease;
rSetupTargets;
bFirstPass:=FALSE;
rOperation;
ENDIF
rLoadConveyor;
rLoadPallet;
rSpecificPart;
ENDPROC

PROC rOperation()
TPReadFK nOperation,"Would you like to Load pallet, Unload pallet or Specify location","Load Pallet",stEmpty,"Unload Pallet",stEmpty,"Specify location";
IF nOperation=1 THEN
rLoadPallet;
ELSEIF nOperation=3 THEN
rLoadConveyor;
ELSEIF nOperation=5 THEN
rSpecificPart;
ELSE
TPWrite "Decision Error!";
ENDIF
ENDPROC

PROC rLoadPallet()
! Loads part to pallet from conveyor
FOR y FROM 1 TO 4 DO
FOR x FROM 1 TO 5 DO
! Get part from conveyor
MoveJ Offs(pConveySim_10,0,0,75),v500,z5,tGripperMain\WObj:=WobjConvey;
MoveL pConveySim_10,v100,fine,tGripperMain\WObj:=WobjConvey;
rGrasp;
MoveL Offs(pConveySim_10,0,0,75),v100,z5,tGripperMain\WObj:=WobjConvey;
! Place part on pallet
MoveJ Offs(pRBPickArray{x,y},0,0,75),v500,z5,tGripperMain\WObj:=WobjRBPallet;
MoveL pRBPickArray{x,y},v100,fine,tGripperMain\WObj:=WobjRBPallet;
rRelease;
MoveL Offs(pRBPickArray{x,y},0,0,75),v100,z5,tGripperMain\WObj:=WobjRBPallet;
ENDFOR
ENDFOR
ENDPROC

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PROC rLoadConveyor()
! Depalletizing to conveyor
FOR y FROM 1 TO 4 DO
  FOR x FROM 1 TO 5 DO
    ! Get the Part from pallet
    MoveJ Offs(pRBPickArray{x,y},0,0,75),v500,z5,tGripperMain\Wobj:=WobjRBPallet;
    MoveL pRBPickArray{x,y},v100,fine,tGripperMain\Wobj:=WobjRBPallet;
    rGrasp;
    MoveL Offs(pRBPickArray{x,y},0,0,75),v100,z5,tGripperMain\Wobj:=WobjRBPallet;
    ! Place part on the conveyor pallet
    MoveJ Offs(pConveySim_10,0,0,75),v500,z5,tGripperMain\Wobj:=WobjConvey;
    MoveL pConveySim_10,v100,fine,tGripperMain\Wobj:=WobjConvey;
    rRelease;
    MoveL Offs(pConveySim_10,0,0,75),v100,z5,tGripperMain\Wobj:=WobjConvey;
  ENDFOR
ENDFOR
ENDPROC

PROC rSpecificPart()
! Picks a specific part from the array/pallet
x:=0;
WHILE x<1 OR x>5 OR x<>Trunc(x) DO
  TReadNum x,"Enter the x position.";
ENDWHILE
y:=0;
WHILE y<1 OR y>4 OR y<>Trunc(y) DO
  TReadNum y,"Enter the y position.";
ENDWHILE
!Get the Part
MoveJ Offs(pRBPickArray{x,y},0,0,75),v500,z5,tGripperMain\Wobj:=WobjRBPallet;
MoveL pRBPickArray{x,y},v100,fine,tGripperMain\Wobj:=WobjRBPallet;
rGrasp;
MoveL Offs(pRBPickArray{x,y},0,0,75),v100,z5,tGripperMain\Wobj:=WobjRBPallet;
! Place part on the conveyor pallet
MoveJ Offs(pConveySim_10,0,0,75),v500,z5,tGripperMain\Wobj:=WobjConvey;
MoveL pConveySim_10,v100,fine,tGripperMain\Wobj:=WobjConvey;
rRelease;
MoveL Offs(pConveySim_10,0,0,75),v100,z5,tGripperMain\Wobj:=WobjConvey;
ENDPROC

PROC rTargetSync()
MoveJ pCylinderBase_10,v100,fine,tGripperMain\Wobj:=WobjRBPallet;
MoveJ pConveySim_10,v100,fine,tGripperMain\Wobj:=WobjConvey;
ENDPROC

PROC rGrasp()
! Close the gripper to grasp a part
WaitRob\InPos;
WaitTime 0.01;
Set doGripActuate;
WaitDI diGripClosed,1;
WaitTime 0.01;
ENDPROC

PROC rRelease()
! Open the gripper to release a part
WaitRob\InPos;
WaitTime 0.01;
Reset doGripActuate;
WaitDI diGripOpen,1;
WaitTime 0.01;
ENDPROC

PROC rSetupTargets()
! Stores target locations in the target array
FOR y FROM 1 TO 4 DO
  FOR x FROM 1 TO 5 DO
    pRBPickArray{x,y}:=Offs(pCylinderBase_10,50*x-50,50*y-50,0);
  ENDFOR
ENDFOR
ENDPROC
ENDMODULE

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